Vision-Based Precision Pose Estimation Using ANN and Particle Filtering

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1. Problem Statement & Motivation

In autonomous formation flight, a trailing follower aircraft must accurately determine its relative position and orientation with respect to a leading aircraft to fly safely and save fuel. While GPS is widely used, it can be jammed or suffer from latency. Relying on an onboard follower camera to track visible features on the leader is a passive, reliable alternative. Direct continuous pose regression from images is notoriously difficult because perspective foreshortening and fuselage self-occlusions create ambiguous, multi-modal solutions. In this project, we reproduce and extend the hybrid framework proposed by Rohan Punnoose (Stanford CS229, 2020). The problem estimates 4 degrees of freedom: Forward separation X, Lateral Y, Vertical Z, and Boresight Roll (phi) (relative pitch and yaw are assumed zero during cruise). We combine an Artificial Neural Network (ANN) that maps 14 visual landmarks to a coarse discrete pose class with a model-based Particle Filter (PF) that refines the estimate to continuous coordinates.

2. Synthetic Aircraft Model & Observation Pipeline

Because the reference paper did not publish 3D landmark coordinates, we built an F-16 scale skeleton (15 m length, 10 m wingspan) with 14 structural landmarks: nose tip, tail exhaust, wingtips, vertical fin tip, stabiliser tips, canopy top, mid-wings, stores, intake lip, and ventral fin. The data generator works as follows:

- **Pinhole Camera:** Placed at relative position (X, Y, Z), gimbaled to point at the leader origin. Landmarks are projected onto a 1280x960 image (focal length f = 1500 px, ~46 deg horizontal field of view).
- **Fuselage Self-Occlusion:** We test line-of-sight against an ellipsoid fuselage (semi-axes [6.0, 0.6, 0.7] m). Points occluded by the fuselage, falling behind the camera, or outside the sensor have coordinates set to (0, 0) and their occlusion flag set to 1. Detected points receive sigma = 0.5 px Gaussian noise and 2% random dropout.
- **Pose Discretization (Table I of Punnoose):** Forward X in [-100, 50] m (10 bins), Lateral Y in [-50, 50] m (10 bins), Vertical Z in [-20, 20] m (8 bins), Roll (phi) in [-45, 45] deg (6 bins). This defines 10 x 10 x 8 x 6 = **4,800 discrete pose classes**.
- **Dataset Scale:** 30 samples are generated per class cell (144,000 total samples, 39.9 MB CSV), enforcing a minimum aircraft safety distance of 10.0 m. All samples verify 100% bijective encode/decode consistency.

3. ANN Coarse Classification Architecture

Following Section IV-D of the paper, we implement a PyTorch feedforward network: **42 inputs** → **Linear(100)** → **ReLU** → **Linear(100)** → **ReLU** → **Linear(4800)** (534,400 parameters). Inputs are 14 landmarks x [u, v, occlusion], centered and scaled by image dimensions ((u - cx)/(W/2), (v - cy)/(H/2)) with occluded coordinates zeroed. Standard z-score normalization is fitted strictly on the training set to prevent leakage. We use an 80/20 stratified split (115,200 train / 28,800 validation samples; exactly 6 validation samples per class) trained for 60 epochs with Adam (lr = 0.001, batch size 256) and Cosine Annealing learning rate scheduling. The network reached its best validation checkpoint at **Epoch 57**.

Metric	Target / Chance	Our Result (Epoch 57)	Physical Cell Center Error
Validation Loss	Cross-Entropy	1.4357	Position Mean: 5.78 m, Median: 5.53 m
Top-1 Exact Class	0.021% (1/4800)	**62.85%** (Paper: ~54%)	Position 90th percentile: 8.77 m
Top-5 Candidate	0.104% (5/4800)	**93.02%**	Roll Mean: 4.16 deg, Median: 3.84 deg
Within +/-1 Bin (All Dims)	Neighbor check	**99.03%** (+/-2 bins: 99.76%)	Per-Dim: X 91.6%, Y 87.9%, Z 82.5%, Roll 92.0%

4. Particle Filter Continuous Refinement

While the ANN identifies the correct pose cell, discretization introduces a physical quantization error floor (cell widths are 15 m in X, 10 m in Y, 5 m in Z, and 15 deg in Roll). We refine this coarse estimate using a model-based particle filter:

- **Initialization:** 2,000 particles are sampled around the top-5 ANN candidate class centers, distributed proportionally to their softmax probabilities with initial spread scaled to cell dimensions.
- **Observation Likelihood:** For each particle, landmarks are projected through our camera model. A robust Gaussian log-likelihood (sigma = 2.0 px, clipped at 8*sigma) scores pixel residual errors on visible landmarks. Mismatches between predicted and observed visibility incur a penalty of 3.0. Stable softmax normalization prevents underflow.
- **Resampling & Jitter Annealing:** Systematic resampling focuses particles on high-likelihood regions. Jitter proposal noise anneals from 30% of cell width down to 4% over 8 iterations, transitioning from wide search to fine continuous convergence. The final pose is the weighted particle expectation.

5. Interactive Demonstration GUI

We developed a clean desktop GUI (src/demo_gui.py) using Python Tkinter. It allows users to set or randomize test poses, enforces safety distance (≥ 10 m), visualizes the 14 projected landmark points on a camera viewport, runs the ANN and particle filter, and displays all three pose estimates (Ground Truth, ANN Coarse, ANN + PF) in a single structured table alongside side-by-side error reductions.

6. Experimental Results & Evaluation Split Audit

In our 100-sample paired evaluation (Phase 4), Particle Filter refinement reduced mean 3D position error from 5.017 m to **0.179 m** (96.43% reduction) and median error from 5.137 m to **0.120 m**. Mean roll error dropped from 4.108 deg to **0.414 deg** (89.92% reduction). Position improved on 99/100 samples and roll improved on 98/100 samples. On 1 sample (Sample 63), the ANN predicted an inverted roll class (+20.5 deg true vs -22.5 deg predicted), trapping the local filter in a secondary minimum (2.16 m → 2.32 m).

Evaluation Split Disclosure: Our audit revealed that the 100 samples in ann_vs_particle_filter.csv were sampled randomly from the full dataset rather than purely from the validation split (81 were from the training set, 19 from validation). While the particle filter never accesses ground truth during estimation, the baseline ANN initialization on those 81 samples reflects training-set performance rather than an independent held-out benchmark.

Metric	ANN Coarse Estimate	ANN + Particle Filter	Measured Improvement
Mean 3-D Position Error	5.017 m	**0.179 m**	**96.43% reduction**
Median Position Error	5.137 m	**0.120 m**	97.67% reduction
90th Percentile Position	7.285 m	**0.332 m**	95.44% reduction
Mean Roll Error	4.108 deg	**0.414 deg**	**89.92% reduction**
Median Roll Error	3.692 deg	**0.069 deg**	98.13% reduction
Refinement Success Rate	Coarse baseline	Pos: 99/100 | Roll: 98/100	1 local minimum outlier

7. Comparison with Reference Paper & Limitations

Key differences between our implementation and Rohan Punnoose (2020) must be clearly stated:

1. **Static Snapshot vs. Flight Trajectory:** Punnoose tested sequential tracking over a 10s flight trajectory with dynamic flight motion and unknown leader disturbances, where his particle filter failed to converge significantly beyond the coarse ANN label. In contrast, our implementation formulates the filter as an iterative static-snapshot optimizer with annealed jitter on independent frames, which converges reliably.
2. **Inverse Simulation:** The particle filter uses the exact same 3D landmark skeleton and projection model used to generate observations. While mathematically sound, this does not model real-world airframe flex, optical distortion, or feature detection noise from raw images.
3. **Feature Detector:** Like the paper, we assume landmark pixel extraction exists as an upstream CNN capability.

8. Conclusions & References

Conclusions: The hybrid ANN + Particle Filter approach successfully bridges discrete classification and continuous pose estimation. The ANN quickly eliminates multi-modal ambiguities by identifying the coarse pose cell, while the particle filter overcomes discretization quantization to reach sub-meter accuracy (0.18 m position, 0.41 deg roll) under the observation model.

References: [1] R. Punnoose, 'Vision-Based Precision Pose Estimation for Autonomous Formation Flying', Stanford CS229 Report, 2020. [2] S. Sharma et al., 'Pose Estimation for Non-Cooperative Spacecraft Rendezvous Using CNNs', IEEE Aerospace, 2018. [3] S. Thrun, 'Probabilistic Robotics', MIT Press, 2005.